

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Endowing large language models (LLMs) with the capacity to actively retrieve external knowledge is essential for complex, real-world tasks whose solutions exceed any model’s static parametric memory. Existing approaches, however, occupy two unsatisfactory extremes: prompting-based methods merely elicit a model’s innate agentic behavior without strengthening it, whereas reinforcement learning (RL) applied to open-ended interactive search is beset by performance ceilings and catastrophic collapse, leaving the underlying agentic potential largely untapped. We resolve both failure modes with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm engineered for robust agent training through the coordinated use of sequence-level optimization and dynamic outcome-based sample filtering. DSPO is trained purely through RL to interleave multi-turn search and reasoning, dispensing entirely with supervised demonstration data. Across seven question-answering benchmarks, our 7B model improves over a comparable prior method by **34.1%**, and, more strikingly, surpasses the 14B model of that prior work on complex multi-hop QA such as HotpotQA by nearly **9% relative**—all while sustaining exceptional training stability.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have attained remarkable performance across a broad spectrum of specialized tasks, spanning mathematics (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code generation (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). Yet a fundamental limitation persists: their knowledge is inherently static, bounded by the corpus on which they were trained. The dominant remedy for this knowledge cutoff is to endow LLMs with search capabilities, transforming them into agents that actively query external knowledge sources (Jin et al., 2025b). This capability constitutes a paradigmatic instance of tool use (Schick et al., 2023), in which the model learns to interact with an external search engine to resolve questions it cannot answer from parametric memory alone. Mastering it demands acquiring a complex, multi-step policy, which naturally frames the problem as sequential decision-making—precisely the setting for which Reinforcement Learning (RL) is designed.

In contrast to Supervised Fine-Tuning (SFT), which depends on costly static demonstrations and cannot teach exploration, RL furnishes a principled framework in which LLMs learn effective policies through trial-and-error interaction with their environment (Chu et al., 2025). Value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have accordingly emerged as the dominant paradigm, prized for their simplicity and markedly reduced memory footprint. Despite their success on more constrained tasks, however, transferring these methods to the open-ended domain of interactive search exposes critical instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). This fragility traces to two fundamental and compounding flaws. First, as established by Zheng et al. (2025), GRPO’s token-level objective is ill-posed when paired with a sequence-level reward: the mismatch produces high-variance gradient estimates that destabilize training. Second, the sparse rewards intrinsic to search tasks frequently yield sample groups with homogeneous outcomes—uniformly successful or uniformly failed—which collapses the advantage signal to zero and starves the optimizer of any learning signal, severely degrading sample efficiency (Yu et al., 2025; Liu et al., 2025).

To confront these two failure modes directly, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an algorithm that synthesizes and refines complementary principles from recent policy optimization research into a single coherent framework. To eliminate the reward–optimization mismatch, DSPO adopts the sequence-level formulation of GSPO (Zheng et al., 2025), matching the unit of optimization to the unit of reward and thereby aligning the objective with the trajectory-level signal that actually governs credit assignment; this alignment fundamentally stabilizes learning over long-horizon reasoning. To eliminate the vanishing-signal problem, DSPO further incorporates a

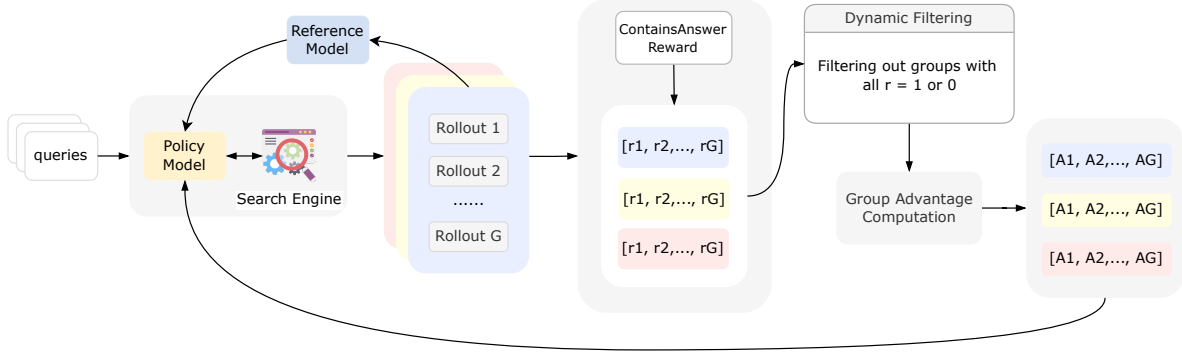


Figure 1: Overview of the DSPO training loop. For a given query, the policy model generates a group of G trajectories by interacting with the search environment, and each trajectory receives a sparse terminal reward. The **dynamic filter** discards groups with homogeneous outcomes and continues sampling until the batch is filled, guaranteeing that every training batch carries an effective, non-degenerate advantage signal. The resulting advantages are then used to update the policy model through the sequence-level objective.

dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025), which constructs each training batch exclusively from rollout groups that contain both successful and unsuccessful trajectories for a given prompt, guaranteeing that the advantage signal $\hat{A}_i$ is well-defined and informative at every update. The two components are mutually reinforcing rather than merely additive: sequence-level updates provide the stability, and dynamic filtering supplies the dense signal, together yielding a stable, high-performance algorithm purpose-built for complex, multi-turn search and reasoning. Trained under this framework, our 7B model attains a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b) and, more strikingly, surpasses its 14B counterpart (Jin et al., 2025a) on complex multi-hop benchmarks such as HotpotQA, exceeding it by **nearly 9% relative** (0.613 vs. 0.563).

Our main contributions are as follows:

- **A unified and stable algorithm for agentic-search RL.** We propose DSPO, which resolves the twin problems of training instability and sample inefficiency that have hindered RL for agentic search by integrating two mutually reinforcing principles within a single cohesive framework: **sequence-level optimization** for robust, low-variance policy updates, and **dynamic outcome-based filtering** for a dense, consistently informative learning signal.
- **State-of-the-art performance verified through rigorous benchmarking.** Evaluated across seven QA benchmarks under a controlled protocol, our DSPO-trained 7B model delivers a **34.1% relative improvement** over a comparable 7B baseline and, more strikingly, outperforms its 14B counterpart on complex multi-hop QA, achieving a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563)—evidence that the gains derive from a superior learning algorithm rather than sheer model scale.
- **Demonstrated training stability under a deliberately minimal retrieval setup.** We provide extensive empirical evidence that DSPO follows a smooth, monotonic learning trajectory where token-level baselines collapse. Because these results are obtained with only a basic BM25 retriever, the observed gains are attributable to the robustness of the algorithm itself rather than to a stronger retrieval component.

2 RELATED WORK

2.1 RL FOR LLMs

The landscape of RL for LLMs has evolved rapidly, progressing from foundational Reinforcement Learning from Human Feedback (RLHF) methods built on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler, direct-optimization frameworks such as DPO (Rafailov et al., 2023). A decisive shift toward value-free optimization was marked by Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which dispenses with a learned value model by deriving the reward signal from group statistics. GRPO’s token-level objective, however, is a well-documented source of training instability (Liu et al., 2025; Cui et al., 2025), and this has prompted a series of targeted refinements: GSPO reformulates the objective at the sequence level to

match the unit of reward (Zheng et al., 2025); DAPO addresses inefficient learning under sparse rewards through a dynamic outcome-based sampling mechanism (Yu et al., 2025); and, in a related vein, GMPO stabilizes the token-level objective via geometric-mean aggregation that suppresses the influence of outliers (Zhao et al., 2025). Each of these advances targets one failure mode in isolation, and in our experiments the corresponding algorithms still succumb to training collapse or plateau at a suboptimal performance ceiling. Building on this line of work, DSPO unifies sequence-level optimization and dynamic outcome-based filtering within a single algorithm, jointly resolving the instability and signal-sparsity that uniquely afflict the training of autonomous search agents.

2.2 LLMs WITH AGENTIC RETRIEVAL

To mitigate the static-knowledge limitation of LLMs, Retrieval-Augmented Generation (RAG) couples the model with external retrievers that dynamically incorporate evolving information (Lewis et al., 2020; Gao et al., 2023). Classic RAG frameworks employ dense retrievers to fetch relevant documents and concatenate them into the LLM’s input for generation (Karpukhin et al., 2020); because such pipelines are fixed, however, they afford little autonomy in complex, multi-turn scenarios. Research has consequently moved toward agentic paradigms in which LLMs act as autonomous agents that plan, search, and reason iteratively. ReAct interleaves reasoning and acting to let LLMs operate tools for tasks such as web navigation (Yao et al., 2023), while multi-agent systems such as AutoGen orchestrate collaborative workflows (Wu et al., 2024). A more recent thrust foregrounds the integration of agentic RAG with RL, wherein agents strengthen retrieval through learned decision-making: Wu et al. (2025) introduce Agentic Reasoning, which integrates external tools for streamlined LLM reasoning, and a family of RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) trains LLMs to interleave reasoning and search through RL alone. This end-to-end paradigm internalizes agentic capability within a single policy and sidesteps the engineering overhead of multi-agent orchestration, yet it continues to be constrained by training instability and performance ceilings in the open-ended search domain. Our work confronts precisely these bottlenecks: DSPO provides a robust and efficient training framework that guarantees stable policy optimization, enabling LLMs to acquire effective multi-turn search strategies.

3 METHODOLOGY

This section develops DSPO from first principles. We first formalize agentic search as an RL problem and review prior policy optimization algorithms, making explicit the limitations that render them inadequate in this setting. We then introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, detailing the two components that respectively secure training stability and training efficiency, and characterizing why each is necessary. Finally, we assemble these components into the integrated training algorithm.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL-based training of LLMs commonly relies on policy gradient methods, of which PPO (Schulman et al., 2017) is the canonical instance for alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective evaluated on samples drawn from an old policy $\pi_{\theta_{\text{old}}}$. Averaged over tokens, the objective is

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. A central drawback of PPO is its reliance on a separately trained value model to estimate token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015), which incurs substantial memory overhead and constitutes an additional source of instability.

GRPO (Shao et al., 2024) was proposed to remove this dependency. Rather than learning a value model, GRPO samples a group of G responses $\{y_i\}_{i=1}^G$ for each prompt x and estimates the advantage of each response by normalizing its reward against the group statistics. As in PPO, the objective is optimized at the token level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x,y_{i,<t})}$, and the advantage assigned to every token $y_{i,t}$ within a response y_i is set to the common sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

This construction is the crux of the mismatch we later exploit: although the reward R_i is defined at the level of the whole sequence, GRPO applies the shared advantage $\hat{A}_i$ through a token-level importance ratio, coupling a sequence-level signal to token-level updates.

Agentic Search as a Markov Decision Process. We formalize the iterative process of agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP), which makes the sequential structure of the task—and hence the locus of the credit-assignment problem—explicit.

- **State (s_t):** At turn t , the state s_t encodes the complete interaction history: the initial question q , every previously generated thought and search query, and all retrieved evidence returned by the environment. Because each retrieval must be interpreted and integrated into subsequent decisions, the state grows monotonically along the trajectory, inducing long-horizon dependencies that conventional token-level RL is ill-equipped to optimize.
- **Action (a_t):** An action is a complete textual segment generated by the policy π_θ , comprising free-form reasoning followed by a decision. It terminates either with a `</tool_call>` token, which issues a search and returns the results within `</tool_response>`, or with a `</answer>` token, which ends the trajectory. This structured action space compels the agent to learn not only *what* to generate but also *when* to search, a coupling that induces substantial variability in trajectory length.
- **Policy (π_θ):** The policy π_θ is the underlying LLM, generating tokens autoregressively conditioned on the state. Although it conditions on the full history, the optimization target is computed only over the model-generated thoughts and actions, with the environment-returned content masked out (Jin et al., 2025b). Optimizing π_θ therefore demands credit assignment across long sequences in which many intermediate reasoning steps receive no direct supervision or reward—a regime that makes stable sequence-level optimization indispensable rather than merely advantageous.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records the full sequence of reasoning and tool interactions for a given question, encompassing both model-generated actions and environment-returned search results. Since the reward is assigned to the trajectory as a whole, the optimization problem is intrinsically sequence-level: every early decision can propagate to the eventual answer’s correctness.
- **Reward ($R(\tau)$):** We adopt a sparse terminal reward that grants $R = 1$ when the final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we treat the search engine $\mathcal{S}$ as an explicit component of the environment, with which the policy LLM π_θ interacts by interleaving reasoning and retrieval calls. The overall objective is to find a policy that maximizes the expected terminal reward, regularized by a KL-divergence term that penalizes large deviations from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x;\mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x;\mathcal{S}) || \pi_{\text{ref}}(y|x;\mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x;\mathcal{S})$ signifies that the trajectory y arises from a multi-step process that couples the policy’s token generation with the evidence returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) have successfully cast agentic search as an RL problem, applying conventional algorithms such as PPO or GRPO within it encounters two structural obstacles, both amplified by the open-ended nature of the search environment. The first is a fundamental mismatch between the unit at which reward is assigned (the sequence) and the unit at which optimization is performed (the token) (Zheng et al., 2025); over long trajectories, this discrepancy produces high-variance gradient estimates that accumulate turn after turn and frequently culminate in policy collapse. The second stems from the sparse binary reward: because many training batches contain exclusively successful or exclusively unsuccessful trajectories, the group-normalized advantage degenerates and conveys no learning signal, drastically curtailing sample efficiency (Yu et al., 2025; Liu et al., 2025). DSPO is designed to neutralize these two failure modes simultaneously, addressing the first through sequence-level optimization and the second through dynamic outcome-based filtering.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO advances two coordinated innovations over prior methods: (1) it performs policy optimization at the sequence level, aligning the training objective with the trajectory-based reward structure and thereby eliminating the reward–optimization mismatch; and (2) it introduces a dynamic filtering mechanism that guarantees every training batch carries a high-quality, non-degenerate learning signal. The full training process integrating these components is depicted in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following the sequence-level formulation of GSPO (Zheng et al., 2025), we replace the unstable token-level importance ratio with a length-normalized sequence-level counterpart. The sequence-level importance ratio $s_i(\theta)$ for a response y_i is defined as the geometric mean of its token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length normalization is essential: by the exponent $1/|y_i|$, it confines $s_i(\theta)$ to a consistent numerical range irrespective of sequence length, suppressing the variance that would otherwise accumulate across long trajectories and rendering the clipping operation stable and well-calibrated.

Gradient Analysis. A comparison of gradients makes the source of DSPO’s stability precise. The (unclipped) gradient of the token-level GRPO objective scales each token’s log-probability gradient by a noisy, token-specific weight $r_{i,t}(\theta)$, so estimation error compounds across the tokens of a long trajectory. The gradient of our sequence-level objective instead scales the aggregate log-probability gradient of the entire sequence by a single, more stable sequence-level weight $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying a single holistic correction factor to the entire trajectory, DSPO eliminates the token-level noise accumulation that destabilizes prior methods, yielding fundamentally more stable training. Complementarily, the dynamic filtering mechanism guarantees a well-defined advantage signal $\hat{A}_i$ by admitting only rollout groups that contain both successes and failures, thereby eliminating the wasted, signal-free samples that pervade sparse-reward environments.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary structure of our reward function poses an acute challenge for group-based advantage estimation. Whenever all G responses in a group share the same outcome—uniformly correct ($R = 1$) or uniformly incorrect ($R = 0$)—the group’s reward variance vanishes, and the normalized advantage $\hat{A}_i$ collapses to zero or becomes undefined. Such groups contribute no useful gradient and squander the computation expended on generating them.

DSPO overcomes this pathology through a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, we retain only those trajectory groups that exhibit a mixture of successful and unsuccessful outcomes; formally, a group $\{y_i\}_{i=1}^G$ is admitted for training if and only if its rewards $\{R_i\}_{i=1}^G$ satisfy

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition ensures that the reward variance within every training group is strictly positive, so the group-normalized advantage is always well-defined and informative. In effect, the filter curates a high-quality batch for each policy update, converting an otherwise sparse reward landscape into a dense and efficient learning signal.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
```

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Integrating the two components yields the complete DSPO objective. For each admitted group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ by group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant added for numerical stability. The policy π_{θ} is then updated by maximizing

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined above as the geometric mean of the token-level ratios. Crucially, during likelihood computation we apply loss masking to every token retrieved from the search tool, following Jin et al. (2025b); this restricts the gradient to model-generated content and ensures the policy learns to *reason over* external knowledge rather than merely to reproduce it. The complete procedure is specified in Algorithm 1.

4 EXPERIMENTS

We now present a comprehensive empirical evaluation designed to substantiate the effectiveness of Dynamic-filter Sequence-level Policy Optimization (DSPO). The experiments are organized around three claims: that DSPO (1) attains state-of-the-art performance on a suite of challenging question-answering benchmarks; (2) exhibits markedly superior training stability, avoiding the catastrophic collapse that afflicts baseline methods; and (3) derives its gains from the synergy of its two components, each of which is shown to be indispensable through controlled ablations.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we adopt the prompt template shown in Table 1, which structures the model’s actions during search into three interleaved components—reasoning within `<think>`, retrieval within `<tool_call>`, and the final response within `<answer>`.

Benchmarks and Baselines. To ensure a rigorous and directly comparable evaluation, our experimental design adheres strictly to the established protocol of Search-R1 (Jin et al., 2025b). We train on a composite dataset comprising

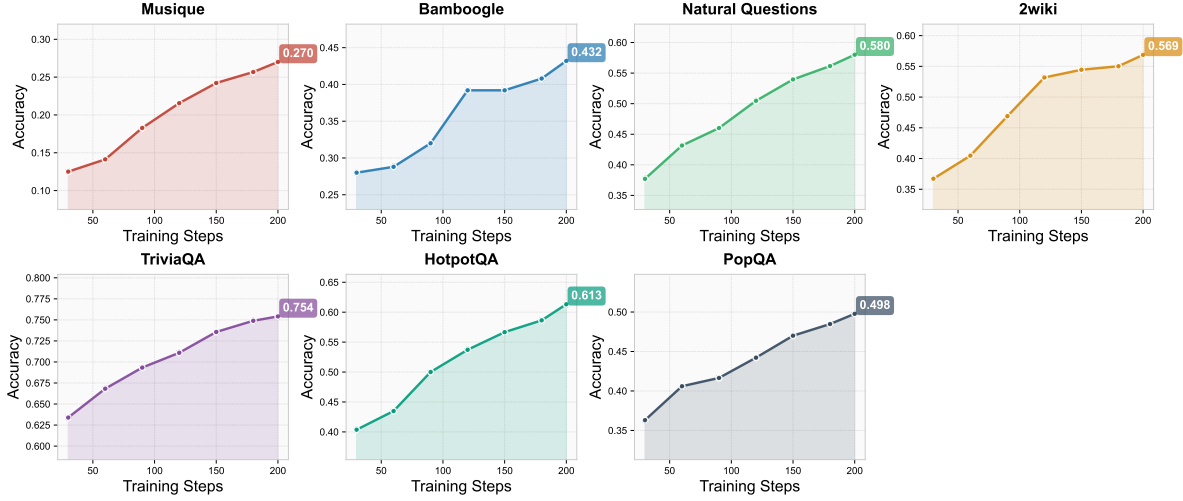


Figure 2: **Validation performance of DSPO across seven benchmarks during training.** Accuracy rises steadily and monotonically on every benchmark, confirming that DSPO’s improvement in training reward transfers directly into stronger generalization rather than reward hacking, and that the method converges to a robust search-and-reasoning policy.

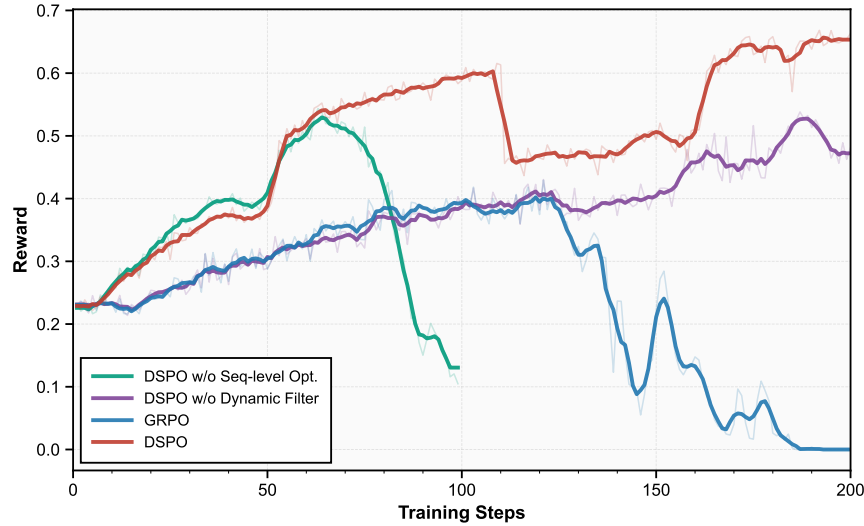


Figure 3: **Training reward dynamics of DSPO and its ablations.** Comparative learning curves reveal a sharp separation by design choice. DSPO (red) improves stably and monotonically to the highest reward, whereas the token-level variants (green, blue) undergo catastrophic policy collapse after a brief initial rise, and the sequence-level variant without the dynamic filter (purple) remains stable but plateaus at a markedly suboptimal level.

the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), and we assess generalization on the test sets of seven diverse QA benchmarks that span single-hop and multi-hop reasoning: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

Our comparison suite pairs strong external baselines with controlled internal ablations, isolating both the overall advantage of DSPO and the contribution of each component. The external baselines are the Qwen2.5-7B and 14B models trained with PPO and GRPO under the Search-R1 framework (Jin et al., 2025b;a). To decompose our method, we further evaluate two ablations that each disable one component: (1) **DSPO w/o dynamic filter**, which reduces to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to the strong token-level policy DAPO (Yu et al., 2025).

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`, first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: The prompt template used in our experiments, which instructs the model to interleave reasoning, retrieval, and answering within the designated tags.

For implementation, all training runs initialize from the Qwen2.5-7B-Instruct checkpoint. Our experiments are built on the VeRL framework (Sheng et al., 2025), adapting its provided Search-R1-like example code and scripts to our methodology. Because our setup employs a modified reward function, we do not simply cite the published Search-R1 numbers; instead, we retrain the PPO and GRPO baselines (Jin et al., 2025b;a) under our exact experimental conditions, and these retrained models constitute our primary external benchmarks. This design guarantees that every reported comparison holds the reward definition, data, and retrieval setup fixed, so differences in performance reflect the learning algorithm alone.

Implementation and Evaluation. To isolate the contribution of the learning algorithm from that of retrieval quality, all RL experiments deliberately employ a standard, unlearned BM25 retriever. This controlled choice ensures that the observed improvements are attributable to the learned policy rather than to a stronger retrieval component. Across all methods, models are trained with a sparse binary reward defined by substring Exact Match (subEM) of the final answer, and subEM likewise serves as the primary evaluation metric, so training signal and evaluation criterion are consistent throughout.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 presents a comprehensive comparison that consolidates our main results and ablation study into a single view. Because DSPO’s two components act synergistically, juxtaposing the full method against both external state-of-the-art baselines and its own ablated variants within one table is the most direct way to convey two distinct conclusions at once: the overall superiority of DSPO relative to prior work, and the indispensability of each of its components.

Table 2: Comprehensive comparison of DSPO against external baselines and ablation variants on seven QA benchmarks, reported as subEM. External baselines are the Search-R1 models (7B and 14B) trained with GRPO and PPO (Jin et al., 2025b;a); the ablations disable one component each (‘w/o dynamic filter’ and ‘w/o seq-level opt.’). For a consistent comparison, all baselines were retrained and re-evaluated under our adjusted reward, with the originally published Search-R1 EM scores shown in parentheses for reference. Best results are in **bold** and second-best are underlined. The DSPO-trained 7B model attains the best or second-best score on every benchmark and the highest average overall.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results in Table 2 establish DSPO’s clear superiority over comparable methods. Our DSPO-trained 7B agent attains an average subEM of 0.531, the best among all evaluated models and a 34.1%

relative improvement over the same-sized Search-R1 (GRPO, 7B) baseline at 0.377. More strikingly, this 7B agent matches or slightly exceeds the average of the far larger Search-R1 14B models under both GRPO (0.530) and PPO, and on the demanding multi-hop HotpotQA benchmark it reaches 0.613 versus the 14B GRPO model’s 0.563—a nearly 9% relative gain. That a 7B policy can equal or surpass a 14B one under an identical retrieval setup is compelling evidence that DSPO’s gains stem from a more effective and stable learning algorithm rather than from an over-reliance on model scale.

Analysis of Ablations. The ablation results in Table 2 demonstrate that both of DSPO’s components are indispensable, and that removing either one accounts for a large, well-localized share of the full method’s performance. Removing the dynamic filter (‘w/o dynamic filter’) is the more damaging: the average subEM collapses from 0.531 to 0.313. Deprived of the filter, the sequence-level objective is repeatedly presented with homogeneous-reward batches and starved of a usable learning signal, which confirms the filter’s role as the mechanism that keeps the advantage signal informative. Ablating sequence-level optimization (‘w/o sequence-level opt.’) is also markedly detrimental, reducing the average to 0.406. Although this token-level variant surpasses the filter-less one, it falls far short of the full model, and—as the next section shows—it is additionally prone to catastrophic training instability. Together these two outcomes establish that the components are complementary rather than redundant: sequence-level updates supply the stability, the dynamic filter supplies the dense signal, and only their combination achieves the full 0.531 average.

Beyond aggregate metrics, DSPO gives rise to sophisticated search behaviors—recognizing irrelevant results, reformulating queries, and verifying answers across multiple turns (see Appendix A for detailed trajectory examples). Notably, all of these behaviors emerge from pure RL training under DSPO, without any supervised demonstration of search strategy.

4.3 ANALYSIS OF TRAINING DYNAMICS

To substantiate our claims regarding stability and efficiency, we examine the training reward dynamics of DSPO alongside its ablations and key baselines. Figure 3 visualizes these trajectories, and the separation among them directly corroborates our two core design choices.

DSPO (red) ascends smoothly and monotonically, converging efficiently to the highest reward level and thereby confirming the stability conferred by the sequence-level objective. The **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—stand in sharp contrast: after a brief initial rise, their rewards collapse catastrophically, a direct manifestation of the high-variance, token-level gradient updates identified in our analysis. **DSPO w/o Dynamic Filter** (purple), which retains sequence-level updates but lacks the dense learning signal, remains stable yet plateaus at a substantially suboptimal ceiling. The three regimes thus dissociate the two components cleanly—sequence-level updates govern whether training remains stable, while the dynamic filter governs how high performance can climb—and only their combination yields both stability and a high asymptote.

To verify that these reward improvements reflect genuine generalization rather than reward hacking, we track validation performance on key benchmarks throughout training. As Figure 2 shows, DSPO’s validation accuracy on NQ, HotpotQA, and the remaining benchmarks rises consistently in lockstep with its reward curve. This tight correspondence between training reward and held-out accuracy confirms that the agent acquires a genuinely generalizable search-and-reasoning policy.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To probe the robustness of our approach beyond the primary 7B setting, we extend the evaluation along two axes: scaling to a larger model, and transfer to an entirely different reasoning domain.

Scalability to Larger Models. We first ask whether DSPO’s stability benefits persist at larger parameter scales by training Qwen2.5-14B-Instruct under an identical protocol. As detailed in Table 3, DSPO scales cleanly: the DSPO-trained 14B model attains an average accuracy of 60.6%, outperforming the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%**, and it improves over that baseline on all seven benchmarks. The advantage is consistent rather than isolated, confirming that DSPO effectively converts additional model capacity into higher accuracy and that its algorithmic benefits compound with, rather than wash out at, larger scale.

Generalization to Mathematical Reasoning. We next assess whether DSPO’s benefits extend beyond multi-turn search to the distinct setting of single-turn mathematical reasoning, evaluating models from both the Qwen2.5 and Qwen3 families. As reported in Table 4, DSPO surpasses GRPO in every configuration—on Math500 with Qwen2.5-Math-7B (0.798 vs. 0.772) and on Olympiad-Bench with Qwen3-4B (0.755 vs. 0.728)—improving across both model

Table 3: Scalability analysis on Qwen2.5-14B-Instruct. The DSPO-trained 14B model surpasses the GRPO-14B baseline on every benchmark; the final column reports the relative gain of DSPO (14B) over GRPO (14B). Best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

families and sizes. Since the sparse-reward and unstable-gradient pathologies that DSPO addresses are not specific to search, these gains indicate that its principles transfer to general reasoning domains rather than being confined to the agentic-retrieval setting.

Table 4: Generalization to single-turn mathematical reasoning across the Qwen2.5 and Qwen3 families. DSPO outperforms GRPO in both configurations; the final column reports the relative gain. Best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

In this work, we resolved the twin problems of training instability and sample inefficiency that have long hindered RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which secures robust training through two mutually reinforcing components: sequence-level optimization that forestalls catastrophic policy collapse, and a dynamic outcome-based filter that converts sparse rewards into a consistently effective learning signal. Across seven QA benchmarks, DSPO not only delivers state-of-the-art performance—improving over a comparable 7B baseline by 34.1% relative and matching or exceeding 14B baselines—but also exhibits markedly superior training stability, and its advantages persist under scaling to 14B and transfer to mathematical reasoning.

By enabling robust training from environmental feedback alone, DSPO offers a practical and efficient blueprint for building capable LLM agents without costly expert demonstrations. This stable foundation opens clear avenues for future work, including the integration of stronger, learned retrievers and the extension of DSPO to complex, multi-tool settings. Moreover, because sparse rewards and unstable policy gradients are by no means unique to search—and as our mathematical-reasoning experiments already begin to demonstrate—we expect DSPO’s principles to deliver comparable gains in domains such as mathematics and code generation, a direction we regard as especially promising. We contend that the two core tenets of DSPO—matching the unit of optimization to the unit of reward, and guaranteeing the density of the learning signal—constitute durable design principles for the next generation of autonomous AI agents.

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A APPENDIX

To illuminate the search strategies acquired through training, Table A.1 presents representative trajectories produced by our model. These examples concretely illustrate three learned competencies: (1) reformulating queries when an initial search returns irrelevant results, (2) verifying information through follow-up searches, and (3) extracting comprehensive answers from retrieved evidence—each emerging from RL training alone.

Table A.1: Representative trajectories illustrating the search-and-reasoning capabilities our model acquires through DSPO training. The model exhibits sophisticated behaviors, including (a) recognition of irrelevant results, (b) query reformulation, and (c) verification through follow-up search.

Example 1: Presidential Succession (*Query Refinement after Failure*)

Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query ."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President

Example 2: University Identification (*Information Verification*)

Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to <u>Cornell University</u> , which is an <u>Ivy League school</u> . However... it is highly likely that the university in question is Cornell. Let me confirm this. "
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University

Example 3: Biographical Information (*Direct Extraction*)

Question	What is John Andrew Martin's occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist</u> , <u>attorney</u> , <u>soldier</u> , and <u>politician</u> . He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician